

financial institutions.

Reputation Risk

Reputation risk is the risk that potential negative publicity regarding a financial institution's business practices could cause a decline in the customer base, costly litigation, or revenue reductions. Reputation risk is relevant to wholesale payment systems because of the large dollar value of the transactions. Errors or fraud can have serious ramifications on the public perception of a financial institution.

Management can mitigate reputation risk for wholesale payment systems by having an effective public relations program, by developing and maintaining strong customer relationships, and by enacting adequate internal controls over all aspects of the wholesale payment system so that errors do not happen in the first place.

Strategic Risk

Credit Risk

Credit risk is the risk that a counter party will not settle an obligation for full value when due. In relation to wholesale payments and securities settlement systems, credit risk can take several forms and have multiple sources. Two of the most important sources, customer daylight overdrafts and settlement risks (including settlement lags, principal risk, and loss-sharing provisions), are discussed below. These forms of credit risk, as well as other forms of credit risk, should be appropriately monitored and managed. Information technology is often critical to such controls.

Customer Daylight Overdrafts

Financial institutions often permit their individual and corporate customers to incur intraday overdrafts by allowing customers to make payments without available balances. In most cases, overdrafts are eliminated with incoming funds transfers from other institutions (or outgoing securities transfers against payment) by the end of the business day.

Financial institutions engaging in this practice are extending credit to their customers. As such, they should monitor the credit position of individual customers; control the amount of intraday credit extended to each customer; and have guidelines to prevent exceeding approved intraday and overnight overdraft limits. These guidelines should include:

- Reviewing customer credit limits and the frequency and scope of internal credit reviews. In the absence of pre-authorized limits, institutions should have a process for management approval of daylight overdrafts. Authorization should be within the lending authority of approving officers.
- Reporting and approval procedures for payments exceeding established credit limits to ensure officers with sufficient lending authority make approvals.
- Reviewing intraday overdrafts incurred for compliance with established limits as well as approval and reporting requirements.
- Reviewing arrangements/agreements regarding collateralization of credit exposures.

To the extent that these guidelines give consideration to projected incoming payments, the financial institution should be aware of the risk that expected payments may not be received when expected. Moreover, as described below, institutions should also consider whether such payments have been or are expected to be made with finality.

Since daylight overdrafts constitute an extension of credit (no matter the period of time involved), financial institutions' credit policies should include provisions for approving and monitoring intraday credit lines to customers. Daylight overdrafts have the potential to become overnight overdrafts or overnight loans, and institutions should also have procedures to determine limits on overnight overdrafts. Both sets of procedures should be similar to loan portfolio credit analysis. Credit policies and procedures should include:

- Analyzing worthiness of all borrowers with amounts outstanding in excess of the credit line.
- Reviewing reporting and approval procedures for overdrafts and settlement credits exceeding established limits.
- Assessing reporting and approval procedures for payments against uncollected funds.

Depending on the creditworthiness of the customer and the nature of the activity, a financial institution should consider requiring customers to advise the institution of anticipated incoming securities transfers. Financial institutions should also consider requiring the customer to pre-fund all such anticipated transfers, with the understanding that any transfers not pre-funded may be reversed. To further mitigate credit risk, management should consider requiring customers to collateralize intraday overdrafts. As mentioned above, the control of customers' daylight overdrafts is one of several elements of the Self-Assessment process set out in the Board's PSR policy (see Appendix E).

Settlement Risk

In addition to the explicit provision of credit via daylight overdrafts, financial institutions also need to control their exposure to settlement risks incurred through the institutions' participation in interbank payment and settlement systems. In general, settlement risk is the possibility that the completion or settlement of individual transactions or settlement at the interbank funds transfer or securities settlement level more broadly, will not take place as expected. In addition to credit risk, settlement risk often includes elements of liquidity risk.

One important source of settlement risk is any time lag between the origination of the payment or securities settlement instructions and final settlement, and discharge of those instructions. This time difference between the delivery of payment details and payment finality, or settlement lag, creates the possibility that sending institutions could fail during the lag or at least not be able to settle their obligations when due, typically at the end of the day. If the receiving bank credits the anticipated payment proceeds to the customer's account before the payment is final, the receiving bank may be exposed to credit risk from the sending bank until final settlement occurs. As long as final settlement has not occurred, any credits or additional payment activity undertaken on the basis of "unsettled" payment messages results in credit risk. For more information, see "Real-time Gross Settlement Systems," Committee on Payment and Securities Settlement Systems, 1997 at [www.bis.org/publ/](http://www.bis.org/publ/cpss22.pdf)

cpss22.pdf. Financial institutions should analyze and control this credit risk as they would any other extension of credit.

Real-Time Gross Settlement systems, such as Fedwire Funds Service, are not subject to this risk as payments credited to a receiver's account are final at the time of receipt. CHIPS payments were previously subject to this kind of risk. However, starting January 2001, CHIPS began sending payment details to the receiving bank only once a payment became final. Receiving banks may, however, request certain details about payments queued for their receipt. Such information should not be treated as a final payment. Securities settlements that occur at DTC may be subject to this kind of risk.

Another form of settlement risk is caused by a time-lag between the final settlement of two sides of a given trade or transaction (e.g., any difference in timing between the payment for and delivery of securities, or for foreign exchange transactions, the final delivery of one currency prior to the other). Such time lags can put the entire principle value of trades at risk. Several payments and securities settlement systems, including DTC and CLS Bank are designed to avoid this risk. Nonetheless, many foreign exchange transactions are not settled in CLS Bank, and may be subject to this type of settlement risk. Securities settlement systems in other countries in which U.S. institutions participate may also be subject to this form of risk.

Finally, many payments and securities settlement systems, especially those that rely on settlement lags, netting, or intraday credit, often implement various forms of risk controls that have important credit risk implications for system participants. In particular, systems often employ various types of loss-sharing and supplemental liquidity requirements in the event of settlement failures or disruptions. Participants in any payment or securities settlement system should understand the risks related to these settlement failure procedures and should be prepared to make any required supplemental payments of loss allocation assessments as described in the system rules. Financial institutions may also pre-arrange to serve as liquidity providers to payment or securities settlement systems in the event of a settlement disruption. These liquidity arrangements may involve committed lines of credit on either a secured or unsecured basis or foreign exchange swap facilities. Institutions should understand the nature of these special

commitments and be well-prepared to act on them.

Liquidity Risk

Liquidity risk involves the possibility that earnings or capital will be negatively affected by an institution's inability to meet its obligations when they come due. Liquidity risk is the risk that the financial institution cannot settle an obligation for full value when it is due (even if it may be able to settle at some unspecified time in the future). Liquidity problems can result in opportunity costs, defaults in other obligations, or costs associated with obtaining the funds from some other source for some period of time. In addition, operational failures may also negatively affect liquidity if payments do not settle within an expected time period. Until settlement is completed for the day, a financial institution may not be certain what funds it will receive and thus it may not know if its liquidity position is adequate. If an institution overestimates the funds it will receive, even in a system with real-time finality, then it may face a liquidity shortfall. If a shortfall occurs close to the end of the day, an institution could have significant difficulty in raising the liquidity it needs from an alternative source.

Systems that postpone a significant portion of their settlement activity (in dollar terms) toward the end of the day, such as CHIPS, may be particularly exposed to liquidity risk. These risks can also exist in RTGS systems such as Fedwire. As mentioned above, systems or markets that pose various forms of settlement risk also pose forms of liquidity risk. CLS Bank, while eliminating the bulk of principle risk through its payment-versus-payment design, retains significant liquidity risk, as funding is made on a net basis, and pay-in obligations may need to be adjusted in the event that a counterparty is unable to fund its obligations. Other systems, including securities settlement systems, may also be subject to liquidity risks.

To manage and control liquidity risk, it is important for financial institutions to understand the intraday flows associated with their customers' activity to gain an understanding of peak funding needs and typical variations. To smooth a customer's peak credit demands, a depository institution might consider imposing overdraft limits on all or some of its customers. Moreover, institutions must have a clear understanding of all of their proprietary payment and settlement activity in each of the payment and securities settlement systems in which they participate.

Legal (Compliance) Risk

Legal/compliance risk arises from an institution's failure to enact appropriate policies, procedures, or controls to ensure it conforms to laws, regulations, contractual arrangements, and other legally binding agreements and requirements. In particular, legal risks can result if a financial institution does not provide adequate attention to the operating circulars, procedures and rules of the payment and settlement systems in which it participates. Similarly, an institution's contractual relationships with customers, counterparties, and vendors must be sound and appropriate to the relevant legal framework(s) such as payment and bankruptcy frameworks. Contracts, among financial institutions, their customers, and counterparties are also important to allocate risk-sharing responsibilities applicable to payments. Finally, an institution must ensure it is in compliance with all applicable Federal and State laws and regulations governing payments activity, including the Bank Secrecy Act, the USA PATRIOT Act, and laws

regarding economic sanctions Appendix D provides details on the general legal framework for payments and securities settlement systems.

Office of Foreign Assets Control (OFAC)

The Office of Foreign Assets Control (OFAC), an agency of the U.S. Treasury, administers a series of laws imposing economic sanctions against targeted hostile foreign countries to further U.S. foreign policy and national security objectives. The U.S. government exercises economic sanctions through trade embargoes, blocked assets controls, travel bans, and other commercial and financial restrictions. The economic sanctions programs of the U.S. government are powerful foreign policy tools. Their success requires active participation and support of every financial institution. The Secretary of the Treasury manages the sanctions for the U.S. The U.S. Government mandates that all financial institutions located in the U.S., overseas branches of U.S. financial institutions, and, in certain instances, overseas subsidiaries of U.S. financial institutions, comply with economic sanctions and embargo programs administered under regulations issued by OFAC. In general, the regulations:

- Block accounts and other assets of countries identified as being a threat to national security by the President of the United States (this always involves accounts and assets of the sanctioned countries' governments; it may also involve nationals of the sanctioned countries). In addition, OFAC also blocks the accounts of individuals on OFAC's Specially Designated Nationals (SDN) listing who may not be associated with a sanctioned country.
- Prohibit unlicensed trade and financial transactions with such countries. U.S. law requires that assets and accounts be blocked when such property is located in the U.S., is held by U.S. individuals or entities, or comes into the possession or control of U.S. individuals or entities. The definition of assets and property is very broad and covers direct, indirect, present, future, and contingent interests. Certain individuals and entities located around the world that are acting on behalf of sanctioned country governments have been identified by the U.S. Treasury and must be treated as if they are part of the sanctioned governments. U.S. banks must block funds transfers that are remitted:
 - By, or on behalf of a blocked individual or entity;
 - To, or through a blocked entity; or
 - In connection with a transaction in which a blocked individual or entity has an interest.

Financial institutions receiving instructions to make a payment that falls into one of these categories are required to execute the payment order and place the funds into a blocked account. Customers cannot cancel or amend a payment order after the U.S. bank has received it. Once assets or funds are blocked, they may be released only by specific authorization from the U.S. Treasury. If OFAC compliance issues are found during an

examination, the examiner should follow up with the bank regulatory agency's compliance area to determine whether the financial institution needs to acquire subject matter expert support.^[12]

Bank Secrecy Act (BSA)

Financial institutions should develop and provide for the continued administration of a program reasonably designed to ensure and monitor compliance with the record keeping and reporting requirements set forth in subchapter II of the Bank Secrecy Act.^[13] The BSA requires a written compliance program that is approved by the board of directors. The board must note the approval in the board minutes. The compliance program must include, at a minimum:

- Provision for a system of internal controls to ensure ongoing compliance;
- Provision for independent testing for compliance to be conducted by institution personnel or by an outside party;
- Designation of an individual or individuals responsible for coordinating and monitoring day-to-day compliance, and
- Provision for training for appropriate personnel.

USA PATRIOT Act

On October 26, 2001, the President signed the Uniting and Strengthening America by Providing Appropriate Tools Required to Intercept and Obstruct Terrorism (USA PATRIOT) Act. The USA PATRIOT Act contains strong measures to prevent, detect, and prosecute terrorism and international money laundering. The provisions of the USA PATRIOT Act that most affect financial institutions are those contained in Title III. Among other things, Title III amends the Bank Secrecy Act and provides the Treasury Department and federal agencies with enhanced authority to combat international money laundering and block terrorist access to the U.S. financial system.

The Act is far-reaching in scope, covering a broad range of financial activities and institutions. One such provision is section 312 - Due Diligence for Correspondent and Private Banking Accounts. Section 312 requires a U.S. financial institution that maintains a correspondent account or private banking account for a non-U.S. person to establish appropriate and, if necessary, enhanced due diligence procedures to detect and report instances of money laundering. Section 312 also describes specific enhanced due diligence standards for U.S. financial institutions that enter into correspondent banking relationships with foreign banks operating under offshore banking licenses or under banking licenses issued by countries that have been:

- Designated as non-cooperative with international anti-money laundering principles by an international body (such as the Financial Action Task Force) with the concurrence of the U.S. representative to that body, or
- The subject of special measures imposed by the Secretary of the Treasury under section 311 of the USA PATRIOT Act.

In addition, section 312 describes minimum anti-money laundering due diligence standards for the maintenance of private banking accounts by U.S. financial institutions for non-U.S. persons. The Treasury Department (Treasury) is authorized to issue regulations implementing section 312. The Act provides that the provisions of section 312 became effective July 23, 2002, whether or not final regulations were in place. Because of the complexity of the issues raised by the proposed rule, Treasury did not promulgate a final rule by July 23, 2002, but rather issued an interim final rule that was effective immediately. The interim final rule requires that insured depository institutions, U.S. branches and agencies of foreign banks, and Edge and Agreement corporations comply with the statutory requirements of section 312.

The interim final rule also provides compliance guidance to financial institutions. This guidance, which is set forth in supplementary information and not as a regulation, indicates what Treasury would consider as "reasonable" due diligence policies and procedures pending the issuance of a final rule. According to Treasury's guidance, these policies and procedures include (1) focusing on accounts that pose the highest risk of money laundering, (2) according priority to those accounts opened on or after July 23, 2002, and (3) complying with existing best practice standards for banks, such as those issued by the Wolfsberg Group in May 2002, the Clearing House in March 2002, and the Bank for International Settlements in October 2001. Treasury noted that it would be reasonable for an institution not to apply every best practice standard if it has a justifiable basis for not adopting a particular practice.

Until Treasury issues a final rule implementing section 312, examiners should make certain covered banking organizations are aware of the specific provisions of the law and have reasonable policies and procedures in place to assure and monitor compliance. Also, in accordance with existing practices concerning anti-money laundering related matters, examiners should ensure that a banking organization is in compliance with the terms of section 312.

Operational (Transaction) Risk

In the context of payment systems, operational risk is the risk of loss resulting from inadequate or failed internal processes, systems, and staff, or from external events. Operational risk includes various physical and information security risks. Institutions should address operational risk by adopting the appropriate internal and operating controls, information security policies and controls, operational resiliency policies and resources, and by comprehensively and effectively auditing these policies, controls, and resources. Institutions should ensure all policies, controls, and resources are sufficient to meet all regulatory expectations.^[14]

Internal and Operational Controls

Management should consider implementing a variety of specific measures to mitigate or limit operational risks, such as authentication and encryption techniques to ensure the authenticity of the payer and payee as well as prevent unauthorized access to information in transit; and edit checks and automated balancing to verify the integrity of the information relative to the payment order and funds transfer transaction. Additional controls include the use of certified tamper resistant equipment, logical access controls to verify transactions, verification of account balances, and the logging of all transactions and attempts to make a transaction.

Additional internal control measures that management should employ to mitigate wholesale payment system risk include:

- Dual custody and separation of duties for critical payment transaction processing and accounting tasks;
- Payment data verification;
- Clear error processing and problem resolution procedures; and
- Confidential and tamper resistant mailing procedures for bankcards and other sensitive material.

The operational controls for funds transfer operations require clearly defined procedures establishing a control environment which provides for the authorization and authentication of transactions. Financial institutions should establish effective operational controls that identify and document:

- The original payment instructions from the corporate or individual customer to the financial institution and other pertinent information (e.g., account officer, branch manager, terminal entry identity, automated interface identification);
- Every transfer point of data for each step of the manual process (e.g., account officer, message receipt, authentication, data entry, and payment release); and
- Every transfer point of data for each step of an automated process (e.g., SWIFT and Telex, message preparation, data entry, and payment release).

Basic internal controls should be in effect to maintain overall integrity for any funds transfer operation. However, depending on the complexity and volume of operations, certain steps may not be applicable for some institutions. Recommended control objectives for a wholesale funds transfer system include:

- Verifying the accuracy and completeness of the outgoing instruction;

- Protecting original instructions from loss or alteration;
- Authenticating the identity and authority of the sender;
- Ensuring collected balances are available and held for the outgoing payments;
- Ensuring the original unaltered outgoing instruction is entered into the internal accounting system;
- Maintaining a physically secure environment; and
- Maintaining appropriate separation of duties for employees involved in the payment process.

Financial institutions should have funds transfer policies and procedures addressing both the processing of funds transfer messages and the related standards for creating and maintaining source documents. Policies and procedures should include documentation describing all interfaces between the funds transfer application and other back office and customer-related banking processes, and should address the controls relating to crediting, debiting, and reconciling customer and institution account balances. Policies and procedures should also document institution specific compliance requirements to address federal and state regulations including OFAC verification procedures.

Audit

A financial institution's internal auditors should conduct periodic independent reviews of the funds transfer operation, including all pertinent internal policies and procedures. An external audit can supplement or replace internal audit procedures. Financial institution audits should verify the effectiveness of the funds transfer control environment and identify funds transfer deficiencies for correction.

Examiners should perform an evaluation of the institution's audit function to determine whether audit activities related to funds transfer operations are comprehensive and effective. Examiners also should review the auditor's opinion of the adequacy of accounting records and internal controls for funds transfer operations. The review of audit procedures should focus on:

- The scope and frequency of the internal funds transfer audit program;
- The effectiveness of audit procedures in determining any control/operating problems disclosed since the previous examination and what corrective measures management has taken;
- Audit work papers to ensure they document adherence to prescribed audit procedures;
- IT audit coverage of new system enhancements and development projects; and
- External audit findings and recommendations.

Information Security

A financial institution's information security program should include an effective risk assessment methodology that includes an evaluation of risks relating to performing high-risk activities such as funds transfer and other payment-related activities. Management should use risk assessments based on a periodic review of high-risk activities to develop effective standards for adequate separation of duties, physical security, and logical access controls based on the concept of "least possible privilege." Refer to the IT Handbook's Information Security Booklet for more detail.

Management should establish logical access controls on the funds transfer application that assign appropriate access levels to staff members working in the wire room or funds transfer operation. Inappropriate access levels provide the opportunity to create and transmit unauthorized funds transfer messages. The risk is greater without adequate separation of duties. Management should ensure no employees have access to more than one assigned user code unless the code is under dual control. Management should configure message verification rights to ensure adequate separation of duties between employees initiating and employees verifying and sending funds transfer messages.

Business Continuity Planning (BCP)

Financial institutions should recognize their role in supporting systemic financial market processes (e.g., inter-bank payment systems and key market clearance and settlement activities) and understand that service disruptions at their institution may significantly affect the integrity of key financial markets. Critical markets include, but are not limited to, the markets for federal funds, foreign exchange, commercial paper, and government, corporate, and mortgage-backed securities. Firms that play significant roles in critical financial markets are those that participate in sufficient volume or value such that their failure to perform critical activities by the end of a business day could present systemic risk (see Appendix F).

In addition, financial institutions should coordinate BCP development and testing with all applicable third parties. All financial institutions, especially those that play a major role in critical financial markets, are expected to have sufficient business continuity plans, commensurate with their funds transfer activities. Financial institutions should also coordinate testing with other industry participants. Refer to IT Handbook's Business Continuity Planning Booklet for more detail.

Vendor and Third-Party Management

Some financial institutions rely on third party service providers and other financial institutions for wholesale payment system products and services either to enhance the services performed in-house or to offer wholesale payment services that are otherwise not cost effective.

Financial institutions should have adequate due diligence processes, appropriate contract provisions, and service provider monitoring procedures to ensure they conduct wholesale payment operations appropriately. Effective monitoring should include the review of select wholesale payment transactions to ensure they are accurate, reliable,